Grade 9 Biology

Illustrated study notes · Life and Living

Cells · Body systems · Digestion · Breathing · Heart and blood · Reproduction

Contents

1. Cells: the microscope, animal and plant cells, specialised cells
2. Levels of organisation and the systems of the human body
3. The digestive system
4. The respiratory (breathing) system
5. The circulatory system: heart, blood vessels and blood
6. Human reproduction: reproductive systems, puberty, the menstrual cycle, pregnancy

These notes follow the standard Grade 9 Natural Sciences (Life and Living) content. Check them against your school's exam guideline.

1. Cells – the building blocks of life

All living things are made of **cells**. A cell is the **smallest unit of life** – the smallest thing that can carry out all the life processes (feeding, breathing, growing, getting rid of waste, reproducing). Most cells are so small that you need a **microscope** to see them.

The light microscope

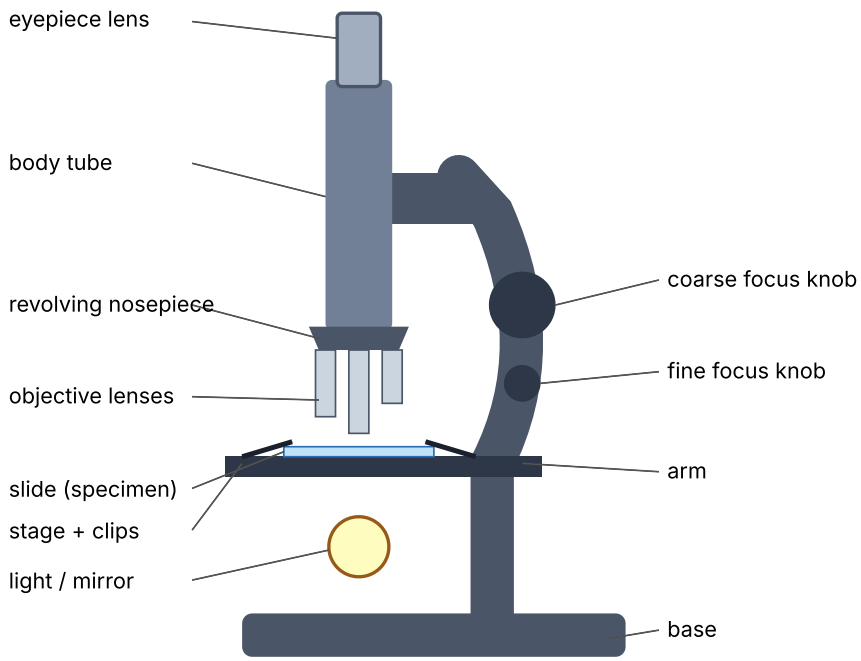


Figure 1: Parts of a light microscope

Part	Function
Eyepiece	Lens you look through; usually magnifies 10×
Objective lenses	Lenses close to the specimen; different magnifications (e.g. 4×, 10×, 40×)
Stage and clips	Flat platform that holds the slide; clips keep it still
Light / mirror	Shines light up through the specimen
Coarse focus knob	Moves the stage a lot – use first, with the lowest-power lens
Fine focus knob	Small movements to make the image sharp
Arm and base	Hold and carry the microscope (one hand on the arm, one under the base)

Total magnification = eyepiece magnification × objective magnification
Eyepiece 10× and objective 40× → 10 × 40 = **400×** (the cell looks 400 times bigger)

The animal cell

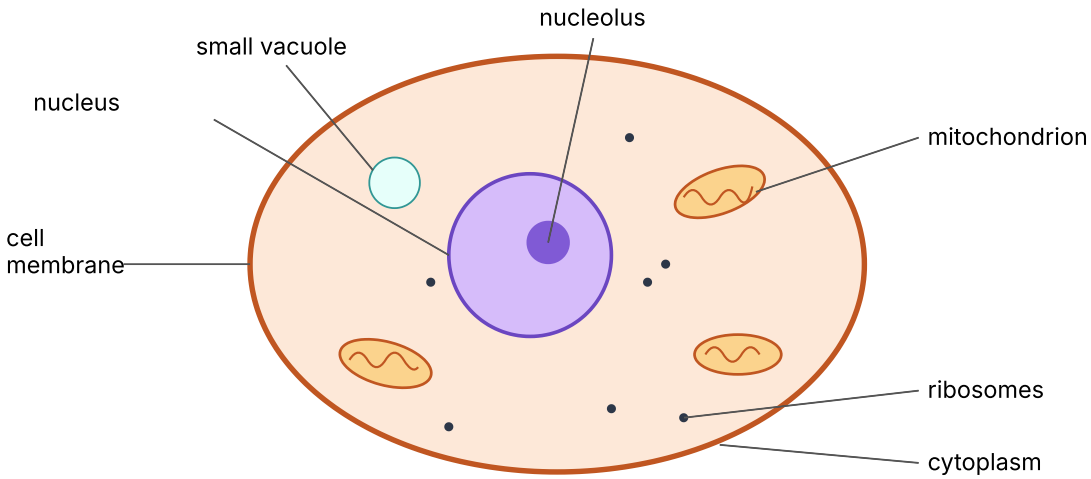


Figure 2: An animal cell (e.g. a cheek cell). Animal cells have no fixed shape.

The plant cell

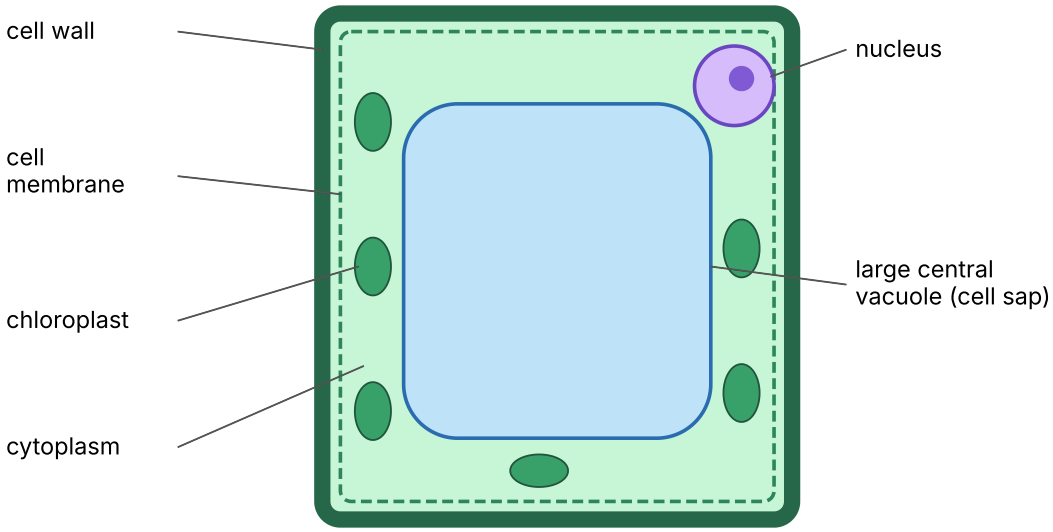


Figure 3: A plant cell (e.g. a leaf cell). The cell wall gives it a fixed, box-like shape.

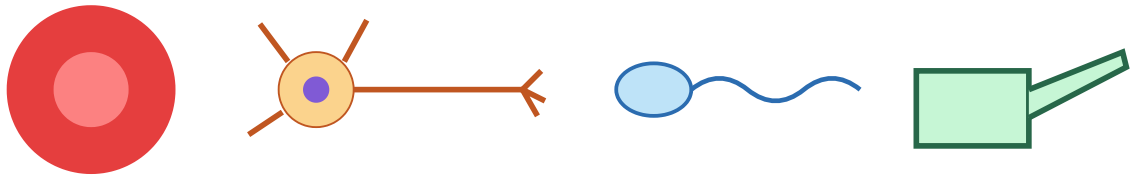
What each part does

Part	Function (job)	Animal cell	Plant cell
Cell membrane	Thin layer around the cell; controls what goes in and out	✔	✔
Cytoplasm	Jelly-like liquid where most chemical reactions happen	✔	✔
Nucleus	The "control centre"; contains the genetic material (DNA / chromosomes)	✔	✔
Mitochondria	Cellular respiration: release energy from food (glucose)	✔	✔
Ribosomes	Make proteins	✔	✔
Cell wall	Strong outer layer made of **cellulose**; gives support and shape	✘	✔
Chloroplasts	Contain green **chlorophyll**; carry out **photosynthesis** (make food)	✘	✔
Vacuole	Stores water and dissolved substances (cell sap); keeps the plant cell firm	small, many / none	one **large** central vacuole

Differences to remember: only **plant cells** have a **cell wall**, **chloroplasts** and a **large central vacuole**. Plant cells have a fixed (regular) shape; animal cells have an irregular shape.

Specialised cells

Cells in a multicellular organism are **specialised** – each type has a special **shape (structure)** that suits its **job (function)**.



Red blood cell
no nucleus, disc shape
→ carries more oxygen

Nerve cell
very long thin fibre
→ carries messages far

Sperm cell
long tail
→ swims to the egg

Root hair cell
long extension
→ absorbs more water

Figure 4: Specialised cells – the shape of each cell suits its job

2. Levels of organisation and body systems

In a multicellular organism, cells are organised into bigger and bigger groups. Each level is made of the level before it.

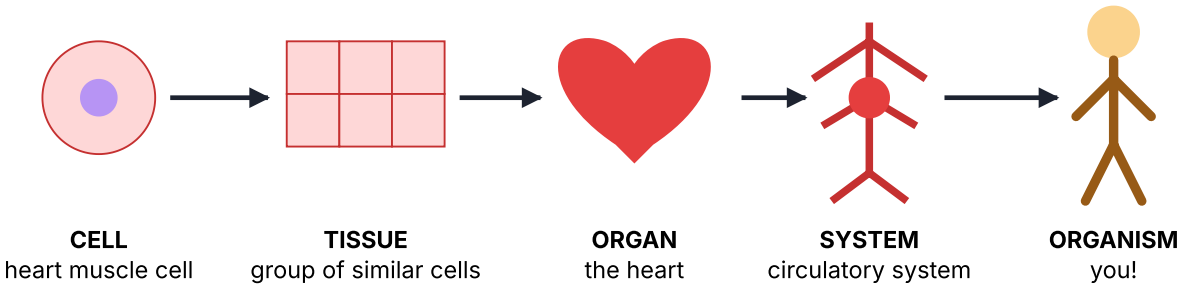


Figure 5: cell → tissue → organ → organ system → organism

- **Tissue:** a group of **similar cells** that do the same job (muscle tissue, nerve tissue, blood).
- **Organ:** different tissues working together to do a job (heart, stomach, lungs, brain).
- **Organ system:** organs that work together for a main function (digestive system).
- **Organism:** all the systems working together make a living thing.

The systems of the human body

> The systems **depend on each other**. Example: the digestive system provides glucose, the respiratory system provides oxygen, and the circulatory system carries both to every cell so the cells can release energy.

System	Main organs	Main job
Skeletal	Bones, joints, cartilage	Support, protection (skull protects the brain), movement, makes blood cells
Muscular	Muscles, tendons	Movement (muscles pull on bones)
Digestive	Mouth, stomach, intestines, liver	Breaks food down so it can be absorbed
Respiratory	Nose, trachea, lungs	Takes in oxygen, removes carbon dioxide (gas exchange)
Circulatory	Heart, blood vessels, blood	Transports oxygen, food, waste and heat around the body
Excretory	Kidneys, bladder, skin, lungs	Removes waste (urine, sweat, CO_2)
Nervous	Brain, spinal cord, nerves	Controls the body; carries messages; reacts to stimuli
Endocrine	Glands (pituitary, thyroid, pancreas, ovaries, testes)	Makes **hormones** – chemical messengers in the blood
Reproductive	Testes / ovaries, uterus	Produces offspring (babies)
Immune	White blood cells, lymph nodes	Fights disease

3. The digestive system

Digestion is the process of **breaking down large food molecules into small, soluble molecules** that can be absorbed into the blood. Food travels through a long tube called the **alimentary canal** (gut), from the mouth to the anus.

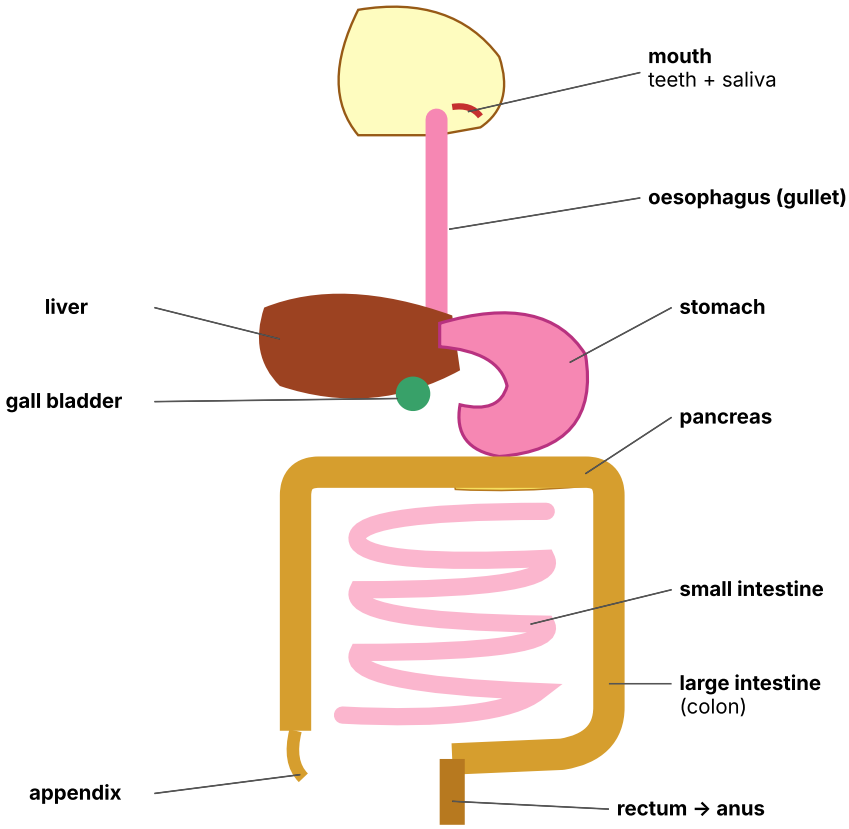


Figure 6: The human digestive system

Two types of digestion

- **Mechanical (physical) digestion:** breaking food into **smaller pieces** – chewing with teeth, churning in the stomach. This gives a bigger surface area for enzymes to work on.
- **Chemical digestion: enzymes** break big food molecules into small molecules. Enzymes are proteins that speed up chemical reactions.

The journey of food

Organ	What happens
Mouth	**Ingestion** (taking in food). Teeth chew (mechanical). Saliva moistens food and contains **amylase**, which starts digesting starch. The tongue rolls food into a ball (**bolus**).
Oesophagus	Muscle tube that pushes food down to the stomach by **peristalsis** – waves of muscle squeezing.
Stomach	Muscular bag that **churns** food. Gastric juice contains **hydrochloric acid** (kills germs) and **pepsin** (starts digesting protein). Food becomes a thick liquid called **chyme**.
Liver and gall bladder	Liver makes **bile**, stored in the gall bladder. Bile breaks fat into small droplets (**emulsification**).
Pancreas	Makes pancreatic juice with enzymes that digest starch, protein and fat. Also makes the hormone **insulin**.
Small intestine	Digestion is completed. **Absorption**: small food molecules pass through the wall into the blood. About 6 m long, lined with millions of **villi**.
Large intestine	**Water** (and some minerals) is absorbed back into the body. Undigested food becomes faeces.
Rectum and anus	Faeces are stored in the rectum and passed out through the anus (**egestion**).

Absorption in the small intestine: villi

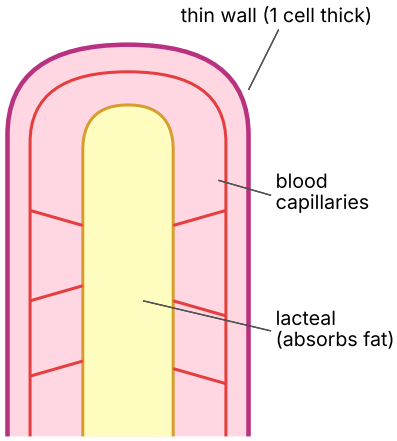


Figure 7: A villus

The inside wall of the small intestine is covered with millions of tiny finger-like folds called **villi** (one is a **villus**). They are perfect for absorption because they:

- give a **very large surface area**,
- have **very thin walls** (one cell thick), so food passes through quickly,
- have a **good blood supply** (capillaries) to carry food away.

Glucose and amino acids go into the **blood**; fats go into the **lacteal**.

Food types and what they are broken into

Food type	Found in	Digested into	Used for
Carbohydrates (starch)	Bread, rice, maize meal, potatoes	**Glucose**	Energy
Proteins	Meat, fish, eggs, beans	**Amino acids**	Growth and repair
Fats (lipids)	Butter, oil, nuts	**Fatty acids + glycerol**	Stored energy, insulation

> **Five stages of digestion:** ingestion (eating) → digestion (breaking down) → absorption (into blood) → assimilation (cells use the food) → egestion (removing undigested waste).

Problems: heartburn (acid moves up the oesophagus), ulcers, constipation (too little fibre and water), diarrhoea (often from germs in dirty water – causes dehydration), obesity and malnutrition. A **balanced diet** with fibre, fruit, vegetables and water keeps the system healthy.

4. The respiratory (breathing) system

The respiratory system brings **oxygen** into the body and removes **carbon dioxide**. Every cell needs oxygen to release energy from glucose (**cellular respiration**).

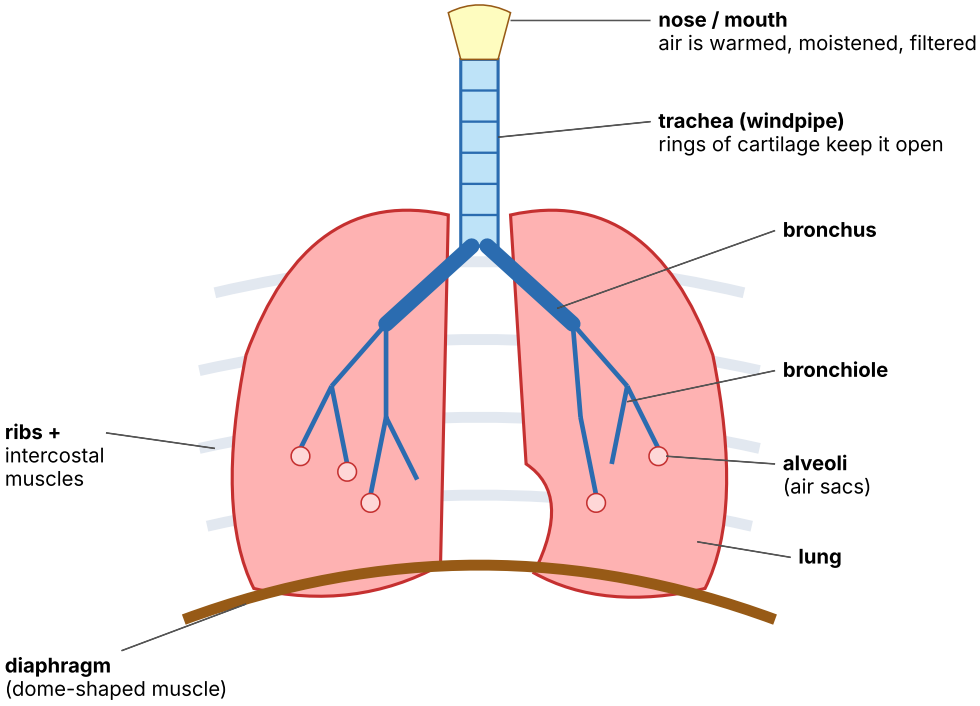


Figure 8: The human respiratory system

The path of air: nose / mouth → **pharynx** → **larynx** (voice box) → **trachea** → **bronchi** (one to each lung) → **bronchioles** → **alveoli**. The **epiglottis** is a flap that closes the trachea when you swallow, so food doesn't go into the lungs.

Gas exchange in the alveoli

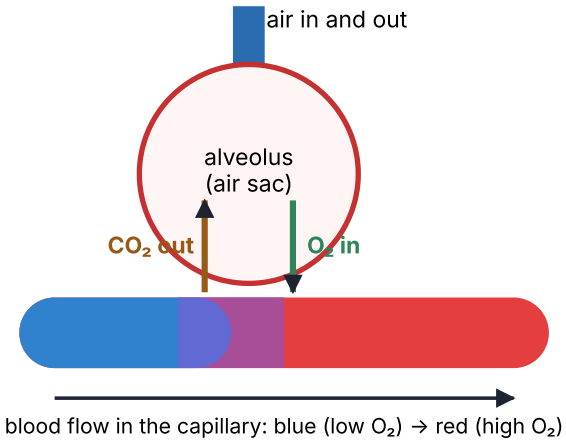


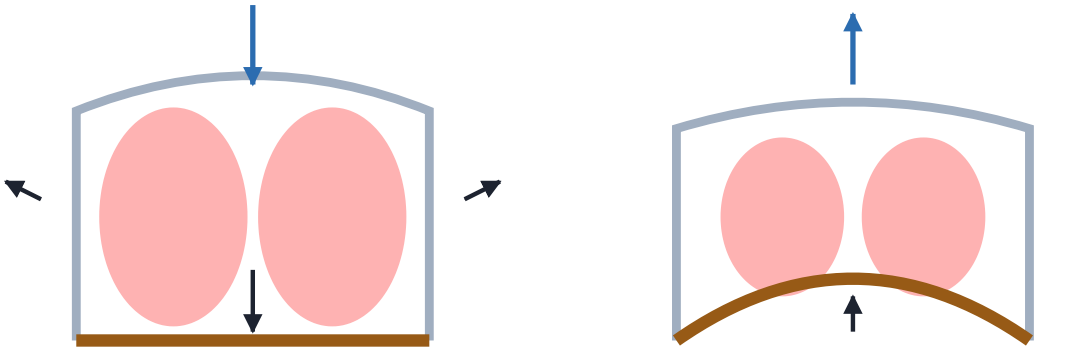
Figure 9: O_2 moves from the air into the blood; CO_2 moves from the blood into the air

Gases move by **diffusion** – from where there is a **lot** of a substance to where there is **less**.

- **Oxygen** diffuses from the air in the alveolus **into the blood**.
- **Carbon dioxide** diffuses from the blood **into the alveolus** and is breathed out.

Alveoli are good at gas exchange because they have: a **huge surface area** (about 300 million alveoli), **very thin walls** (one cell thick), a **moist lining**, and a **rich blood supply**.

Breathing in and out



BREATHING IN (inhalation)
diaphragm flattens, ribs move up and out

BREATHING OUT (exhalation)
diaphragm domes up, ribs move down and in

Figure 10: How the diaphragm and ribs change the size of the chest

	Breathing in (inhalation)	**Breathing out (exhalation)**
Diaphragm	Contracts and **flattens** (moves down)	Relaxes and **domes up**
Intercostal muscles / ribs	Contract; ribs move **up and out**	Relax; ribs move **down and in**
Volume of chest	**Increases**	**Decreases**
Pressure inside the lungs	**Decreases** → air rushes **in**	**Increases** → air is pushed **out**

Gas	**Inhaled air**	**Exhaled air**
Oxygen	about 21%	about 16%
Carbon dioxide	about 0.04%	about 4%
Water vapour	less	more (that's why breath mists a mirror)

> **Breathing ≠ respiration.** **Breathing** is moving air in and out of the lungs. **Cellular respiration** happens in the mitochondria of every cell: **glucose + oxygen → carbon dioxide + water + energy**.

Diseases: asthma (airways narrow, hard to breathe), **tuberculosis (TB)** (bacteria infect the lungs; spreads through coughing), **pneumonia**, **bronchitis**, **lung cancer** and **emphysema** (alveoli are damaged – mainly caused by smoking).

5. The circulatory system

The circulatory system is the body's **transport system**. The **heart** pumps **blood** through **blood vessels** to carry oxygen, food (glucose), water, hormones and heat to every cell, and to take carbon dioxide and other waste away.

The heart

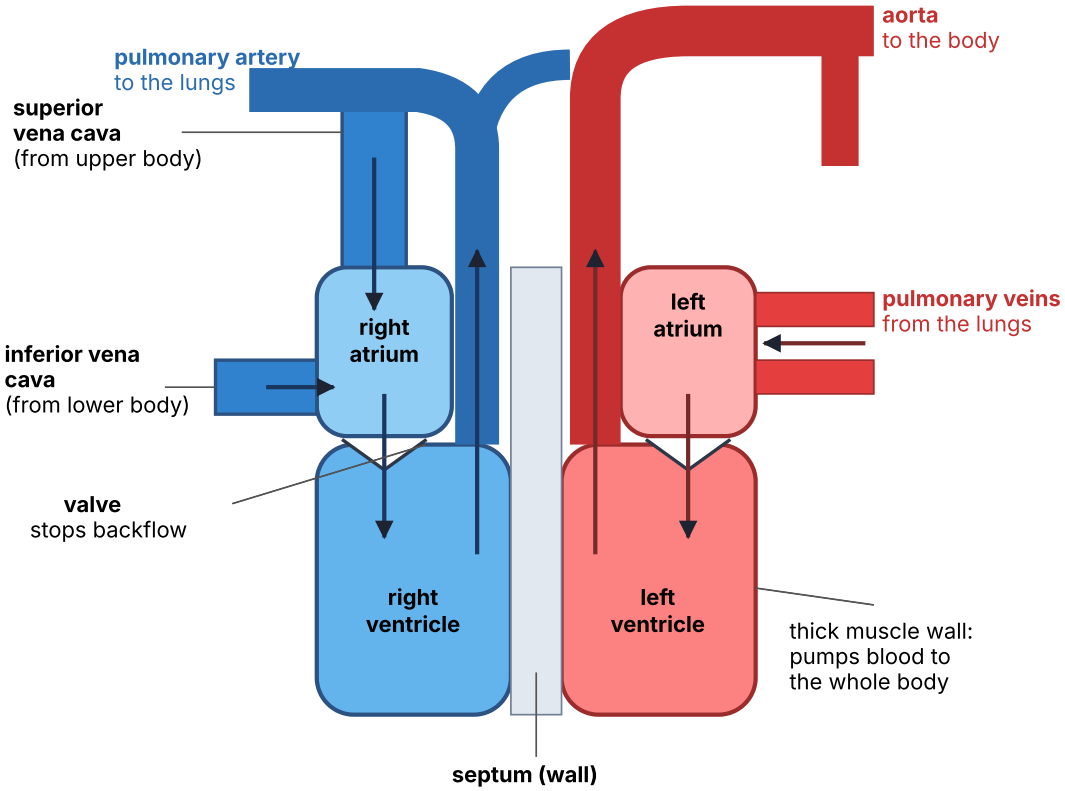


Figure 11: The heart, seen from the front. The heart's left side is on the right of the drawing, as if you are facing the person.

- The heart has **4 chambers**: two **atria** at the top (receive blood) and two **ventricles** at the bottom (pump blood out).
- The **right side** pumps **deoxygenated** blood to the **lungs**. The **left side** pumps **oxygenated** blood to the **whole body**.
- The **left ventricle** has the **thickest wall** because it must pump blood all around the body.
- The **septum** separates the two sides so oxygenated and deoxygenated blood don't mix.
- **Valves** make sure blood flows in **one direction only** (the "lub-dub" sound is valves closing).
- The heart is made of **cardiac muscle**, which never gets tired. Its own blood supply comes from the **coronary arteries**.

Double circulation: the path of blood

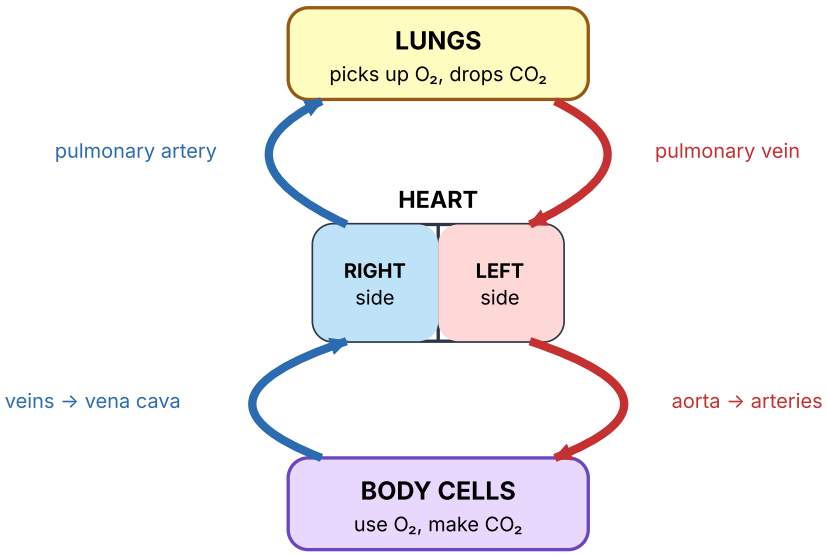


*Figure 12: Blood passes through the heart **twice** on every trip around the body*

> **Learn this order:** body → vena cava → **right atrium** → **right ventricle** → pulmonary artery → **lungs** → pulmonary vein → **left atrium** → **left ventricle** → aorta → body.

- **Pulmonary circulation:** heart → lungs → heart (to pick up oxygen).
- **Systemic circulation:** heart → body → heart (to deliver oxygen and food).

Blood vessels

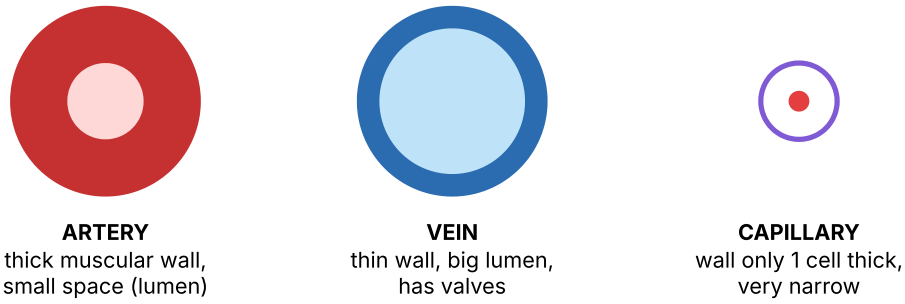


Figure 13: Cross-sections of the three types of blood vessels

	Arteries	Veins	Capillaries
Direction	**Away** from the heart (A for Away)	**Towards** the heart	Connect arteries to veins, between the cells
Blood	Oxygenated (except pulmonary artery)	Deoxygenated (except pulmonary vein)	Gives O_2 and food to cells, takes waste
Pressure	High – the pulse you feel	Low	Low
Wall	Thick, muscular, elastic	Thinner; valves stop backflow	One cell thick – substances pass through

What blood is made of

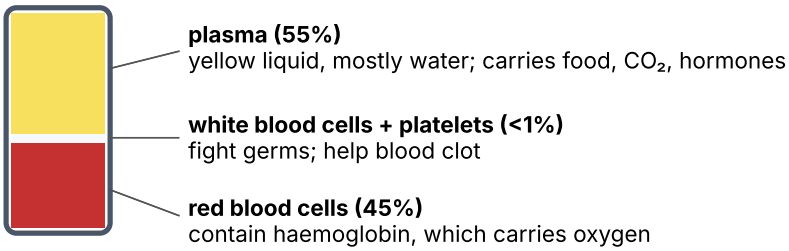


Figure 14: Blood separated in a test tube

Part	Job
Red blood cells	Carry **oxygen** using **haemoglobin** (contains iron). No nucleus, disc-shaped.
White blood cells	**Fight infection**: engulf germs or make antibodies.
Platelets	Help blood **clot** to stop bleeding.
Plasma	Liquid that carries dissolved food, CO_2, waste, hormones and heat.

Diseases and health: high blood pressure (hypertension), **heart attack** (a coronary artery is blocked by fatty deposits), **stroke** (blood supply to the brain is blocked), **anaemia** (too few red blood cells or too little iron – tired and pale). Keep your heart healthy: exercise, eat less fat, salt and sugar, don't smoke, keep a healthy weight.

6. Human reproduction

Reproduction is how living things produce offspring. Humans reproduce **sexually**: a male **sex cell (sperm)** joins a female **sex cell (egg / ovum)**. Sex cells are made in the **gonads** – the **testes** in males and the **ovaries** in females.

Puberty

Puberty is the stage (usually between 9 and 16 years) when the body becomes able to reproduce. It is controlled by **hormones**: **testosterone** in boys and **oestrogen** and **progesterone** in girls.

Changes in boys	Changes in girls	Both
Voice deepens Facial and chest hair Shoulders broaden, muscles develop Testes start making sperm Penis and testes grow	Breasts develop Hips widen Ovaries start releasing eggs Menstruation (periods) begins	Growth spurt Pubic and underarm hair Sweat more, body odour Pimples (acne) Emotional changes, mood swings

The male reproductive system

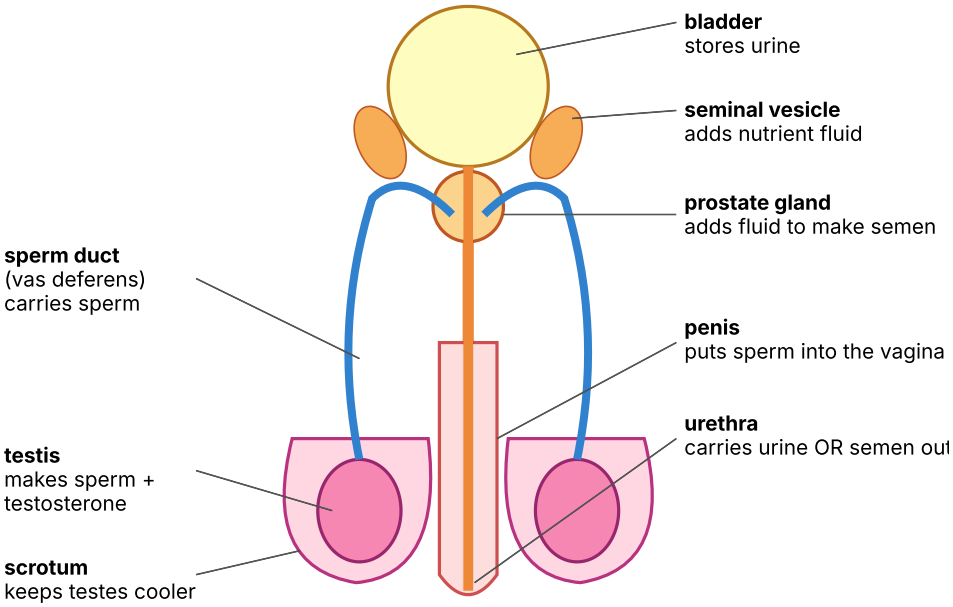


Figure 15: The male reproductive system (front view, simplified)

Path of sperm: testis → sperm duct → (seminal vesicle and prostate add fluid → **semen**) → urethra → out through the penis. Testes hang outside the body in the scrotum because sperm develop best slightly **below body temperature**.

The female reproductive system

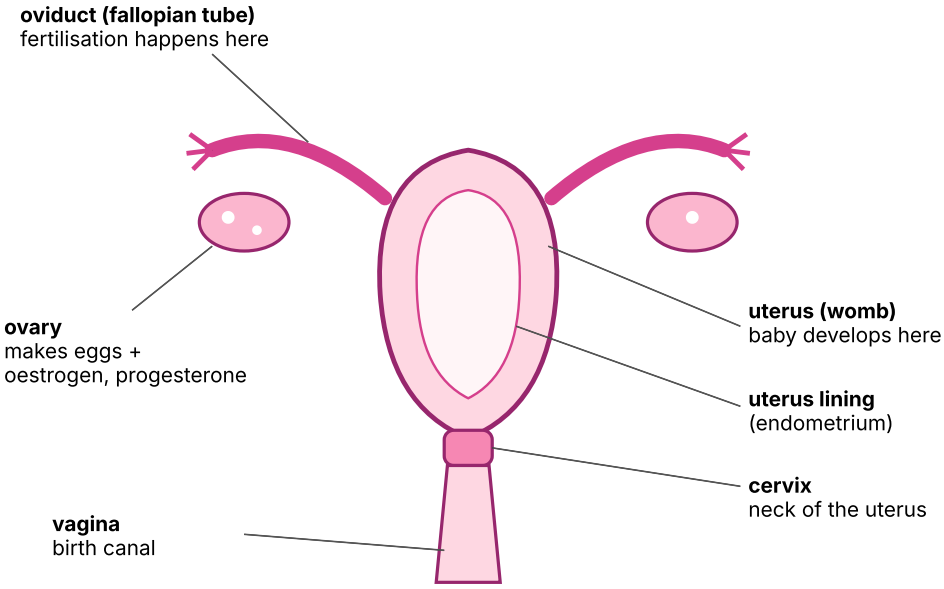


Figure 16: The female reproductive system (front view, simplified)

The menstrual cycle

About every **28 days** the female body prepares for a possible pregnancy. This is the **menstrual cycle**.

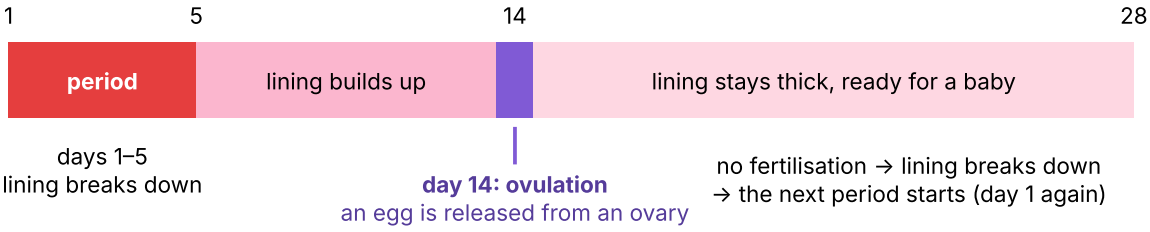
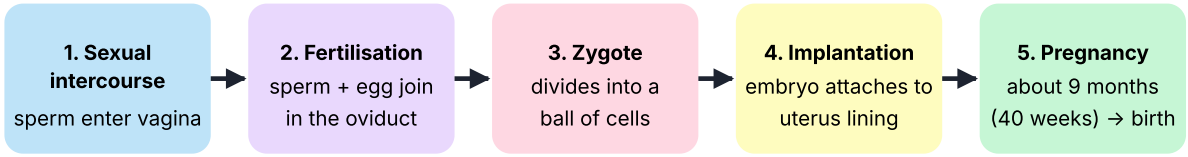


Figure 17: The 28-day menstrual cycle

1. **Days 1–5 – menstruation (period):** the thick lining of the uterus breaks down and passes out through the vagina with some blood.
2. **Days 6–13:** the uterus lining **thickens** again (controlled by oestrogen). An egg ripens in an ovary.
3. **About day 14 – ovulation:** a mature egg is released from an ovary into the oviduct. This is when a woman is most likely to fall pregnant.
4. **Days 15–28:** the lining stays thick (progesterone). If the egg is **not fertilised**, it dies, the lining breaks down and a new cycle starts. If it **is** fertilised, there is no period – the woman is pregnant.

Fertilisation, pregnancy and birth



embryo (first 8 weeks) → foetus (from about 9 weeks until birth)

Figure 18: From fertilisation to birth

- **Fertilisation:** one sperm joins the egg in the **oviduct**. The fertilised egg is a **zygote**. It contains half its genetic material from the father and half from the mother.
- The zygote divides many times as it moves to the uterus and **implants** in the thick lining.
- The **placenta** forms: it lets food and oxygen pass from the mother's blood to the baby, and waste from the baby to the mother. The baby is joined to the placenta by the **umbilical cord**. The **amniotic fluid** protects the baby from bumps.
- Alcohol, smoking and drugs pass through the placenta and can harm the baby (e.g. foetal alcohol syndrome).
- **Birth:** the uterus muscles contract, the cervix opens and the baby is pushed out through the vagina.

Preventing pregnancy and STIs

Method	How it works
Abstinence	Not having sex – the only 100% effective method against pregnancy and STIs
Condom (male / female)	Barrier that stops sperm reaching the egg; also protects against STIs and HIV
Contraceptive pill / injection / implant	Hormones stop ovulation; do **not** protect against STIs
IUD (loop)	Small device in the uterus that stops implantation

Sexually transmitted infections (STIs) spread through sexual contact: HIV, syphilis, gonorrhoea, chlamydia, herpes. Many have no symptoms at first. Prevent them with abstinence, condoms and testing; treat them early at a clinic.

> **Teenage pregnancy** can lead to dropping out of school, health risks for mother and baby, and money problems. Delay sex, get correct information, and talk to a trusted adult or clinic.